

How to Track File's Data

Disk management

- Need to keep track of where file contents are on disk
- Must be able to use this to map byte offset to disk block
- Structure tracking a file's blocks is called an index node or **inode**
- Inodes must be stored on disk, too

Things to keep in mind while designing file structure

- Most files are small
- Much of the disk is allocated to large files
- Many of the I/O operations are made to large files
- Want good sequential and good random access (what do these require?)

Different File Systems

- **Straw Man** (Pintos otherwise not used in practice)
- **FAT** File Allocation Table (DOS and old Windows)
- **FFS** Unix Fast File System (all modern operating systems)